

# FreshTrack AI: An Intelligent Multi-Task Deep Learning System for Fruit Freshness Detection, Quality Grading, and Shelf-Life Prediction

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**Abstract**—Fruit spoilage represents a critical global challenge, with approximately one-third of all food produced for human consumption lost to waste annually. This research presents *FreshTrack AI*, an intelligent multi-task deep learning system designed to address this problem through automated fruit freshness detection, quality grading, and shelf-life prediction from a single image. The system employs EfficientNet-B0 as the backbone architecture (5.3 million parameters, 1280-dimensional feature representations) with four specialized prediction heads: freshness classification (4 classes), quality grading (3 classes), shelf-life regression, and rotation prediction for data augmentation robustness. We introduce a novel out-of-distribution (OOD) detection mechanism that identifies non-fruit inputs through entropy-based uncertainty quantification, achieving robust rejection of unrecognized objects. The system is deployed through two complementary interfaces: a FastAPI-based backend with SQLite prediction logging and a Flutter mobile application supporting both online inference and offline capabilities. Experimental results demonstrate 77.1% accuracy on the freshness classification task with 45 ms inference latency, achieving an optimal balance between accuracy and computational efficiency. Our contributions include: (1) a unified multi-task architecture for comprehensive fruit quality assessment, (2) an entropy-based OOD detection pipeline for unreliable prediction handling, (3) a privacy-compliant mobile deployment framework, and (4) a feedback-driven active learning system for continuous model improvement.

**Index Terms**—Fruit Freshness Detection, Multi-Task Learning, Deep Learning, EfficientNet, MobileNet, Out-of-Distribution Detection, Transfer Learning, Computer Vision, Mobile Deployment

## I. INTRODUCTION

### A. Background on Food Waste and Freshness Detection

The global food waste crisis represents one of the most significant challenges facing modern society. According to the Food and Agriculture Organization (FAO) of the United Nations, approximately 1.3 billion tons of food are lost or wasted annually, representing nearly one-third of all food produced for human consumption [1]. This equates to roughly \$1 trillion in economic losses each year, making food waste not just an environmental concern but a significant economic problem.

Fruits, being perishable commodities with high moisture content, are particularly susceptible to spoilage throughout the

supply chain—from harvest through storage, distribution, and retail. Postharvest losses for fruits and vegetables can range from 20% to 50% in developing countries due to inadequate storage and handling facilities. In developed countries, significant waste occurs at the retail and consumer levels despite lower supply-chain losses.

The economic impact is substantial, with annual losses in the fruit and vegetable sector estimated at over \$100 billion globally. Beyond economic costs, food waste has significant environmental implications, contributing to greenhouse gas emissions when organic matter decomposes in landfills. The UN Sustainable Development Goal 12.3 calls for halving food waste at the retail and consumer levels by 2030.

Traditional fruit quality assessment relies heavily on human inspection, presenting several limitations: (1) *labor intensity*, (2) *subjective inconsistency*, (3) *limited scalability*, (4) *environmental constraints*, and (5) *high operational costs*.

### B. Limitations of Existing Solutions

Current fruit freshness detection systems suffer from:

**Single-Task Focus:** Most existing systems address only one aspect of fruit quality—either freshness classification or quality grading—without providing a comprehensive assessment framework.

**Lack of Uncertainty Quantification:** Traditional classification systems provide overconfident predictions, leading to unreliable behavior when presented with non-fruit inputs.

**Limited Deployment Flexibility:** Many solutions require GPU-accelerated servers, lacking mobile-friendly offline inference capabilities needed in agricultural field settings.

**Absence of Continuous Learning:** Static models cannot adapt to new fruit varieties or evolving visual characteristics driven by climate change or agricultural practices.

### C. Research Contributions

This paper presents FreshTrack AI with four key contributions:

- 1) **Unified Multi-Task Architecture:** A single EfficientNet-B0-based model with four specialized prediction heads for freshness classification (4 classes),

quality grading (3 classes), shelf-life regression, and rotation auxiliary prediction.

- 2) **Entropy-Based OOD Detection:** A novel mechanism identifying non-fruit inputs through Shannon entropy computation, rejecting inputs exceeding entropy thresholds ( $>1.5$  bits) or below confidence thresholds ( $<60\%$ ).
- 3) **Privacy-Compliant Mobile Deployment:** A Flutter-based mobile application with TensorFlow Lite support for offline inference, enabling privacy-preserving fruit assessment without cloud connectivity.
- 4) **Feedback-Driven Active Learning:** A continuous improvement pipeline where user corrections are logged and used for periodic model retraining.

## II. RELATED WORK

### A. Traditional Fruit Quality Assessment

Before deep learning, fruit quality assessment relied on visual inspection by trained experts, near-infrared spectroscopy (NIRS), electronic nose systems using chemical sensor arrays, and hyperspectral imaging for detecting internal defects across multiple wavelengths [2]. While accurate, these approaches are prohibitively expensive or require controlled environments unsuitable for field deployment.

### B. Deep Learning Approaches in Computer Vision

Key CNN architectures applied to fruit quality tasks include:

**ResNet** [2]: Skip connections enable training of very deep networks without gradient degradation, widely applied to fruit classification.

**EfficientNet** [3]: Compound scaling uniformly scales depth, width, and resolution. EfficientNet-B0 achieves comparable accuracy to larger networks with significantly fewer parameters.

**MobileNet** [4]: Depthwise separable convolutions provide efficient inference on mobile devices with excellent accuracy-speed tradeoffs.

### C. Multi-Task Learning in Computer Vision

Multi-task learning (MTL) enables shared representation learning across related tasks, improving efficiency and generalization [5]. Applications include object detection with classification [6], pose estimation with segmentation, and face detection with landmark localization. Zhang et al. [7] proposed a multi-task depthwise separable convolutional network for simultaneous freshness detection and fruit type classification.

### D. Fruit Freshness Detection Literature

Aksoy et al. [8] evaluated four pretrained CNN models for fresh vs. rotten fruit classification, with ShuffleNetV2\_x1\_5 achieving the highest overall accuracy of 94.61%. Lightweight MobileNetV3 Large achieved 91.04% accuracy for avocado ripeness classification with only 26.52 MB memory usage, demonstrating suitability for resource-constrained devices.

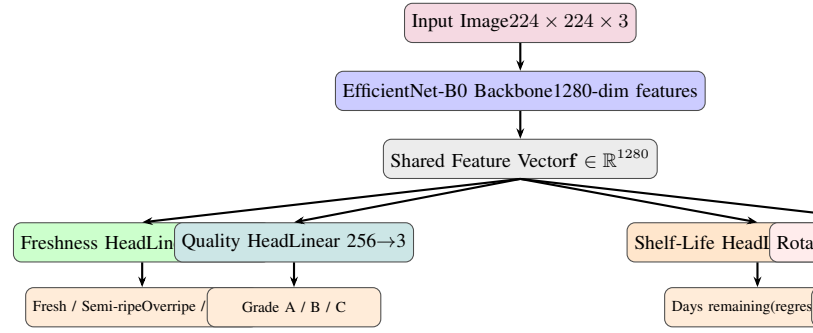


Fig. 1: FreshTrack AI multi-task model architecture. A shared EfficientNet-B0 backbone feeds four specialized prediction heads, enabling joint optimization across all tasks.

### E. Out-of-Distribution Detection

Detecting inputs outside the training distribution is crucial for reliable deployment. Key approaches include confidence thresholding [9], entropy-based methods [10], and Monte Carlo Dropout for Bayesian uncertainty approximation. Our system combines confidence thresholding with entropy-based detection for robust OOD identification.

## III. METHODOLOGY

### A. Model Architecture

FreshTrack AI employs EfficientNet-B0 as a shared backbone encoder followed by four task-specific prediction heads, as illustrated in Fig. 1. The backbone extracts a 1280-dimensional feature vector from  $224 \times 224$  RGB input images. Each head independently processes this shared representation:

- **FreshnessHead:** Dropout(0.3)  $\rightarrow$  Linear(1280 $\rightarrow$ 512)  $\rightarrow$  ReLU  $\rightarrow$  Dropout(0.2)  $\rightarrow$  Linear(512 $\rightarrow$ 4), predicting {Fresh, Semi-ripe, Overripe, Rotten}.
- **QualityHead:** Dropout(0.3)  $\rightarrow$  Linear(1280 $\rightarrow$ 256)  $\rightarrow$  ReLU  $\rightarrow$  Linear(256 $\rightarrow$ 3), predicting grades {A, B, C}.
- **ShelfLifeHead:** Dropout(0.3)  $\rightarrow$  Linear(1280 $\rightarrow$ 256)  $\rightarrow$  ReLU  $\rightarrow$  Linear(256 $\rightarrow$ 128)  $\rightarrow$  ReLU  $\rightarrow$  Linear(128 $\rightarrow$ 1), regressing shelf-life in days.
- **RotationHead:** Linear(1280 $\rightarrow$ 4), predicting rotation  $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$  as an auxiliary self-supervised task.

Total parameter count:  $\approx 5.3\text{M}$  (backbone) +  $\approx 0.8\text{M}$  (heads) = **6.1M parameters**.

### B. Loss Function

The total training loss is a weighted combination of individual task losses:

$$\mathcal{L}_{\text{total}} = 0.4 \mathcal{L}_f + 0.3 \mathcal{L}_q + 0.25 \mathcal{L}_s + 0.05 \mathcal{L}_r \quad (1)$$

where  $\mathcal{L}_f$ ,  $\mathcal{L}_q$ ,  $\mathcal{L}_r$  are cross-entropy losses for freshness (4 classes), quality (3 classes), and rotation (4 classes) respectively, and  $\mathcal{L}_s$  is MSE loss for shelf-life regression. Higher weight on freshness reflects its primary importance in the application context.

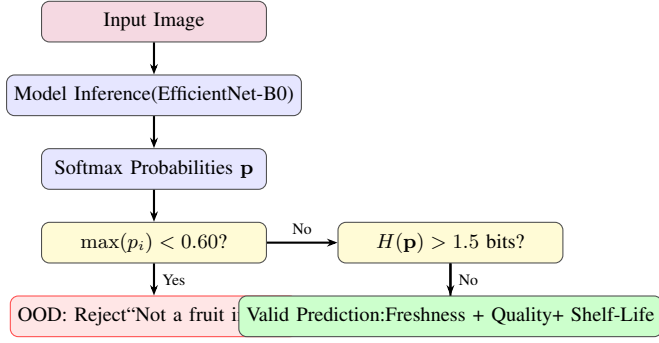


Fig. 2: OOD detection pipeline. Dual-threshold mechanism using maximum confidence and Shannon entropy ensures robust rejection of non-fruit inputs.

### C. Training Configuration

All models were trained on a machine with NVIDIA GPU acceleration using AdamW optimizer with weight decay  $10^{-4}$ . The base EfficientNet-B0 model used a 70/15/15 train/val/test split, batch size 32, learning rate  $10^{-4}$ , and CosineAnnealing-WarmRestarts scheduler over 15 epochs. Full training configuration for all model variants is summarized in Table I.

TABLE I: Training Configuration for Model Variants

| Parameter  | B0 (Base) | B2 (Accuracy) | MobileNet (Speed) |
|------------|-----------|---------------|-------------------|
| Split      | 70/15/15  | 60/20/20      | 80/10/10          |
| Batch size | 32        | 24            | 32                |
| Image size | 224×224   | 260×260       | 224×224           |
| Epochs     | 15        | 15            | 12                |
| LR         | $10^{-4}$ | $10^{-4}$     | $10^{-4}$         |

### D. Data Augmentation

Training images were augmented using the Albumentations library with random horizontal flip ( $p = 0.5$ ), rotation ( $\pm 15^\circ$ ,  $p = 0.5$ ), color jitter (brightness, contrast, saturation  $\pm 0.2$ ), followed by ImageNet normalization. These augmentations improve generalization across varying lighting conditions and orientations encountered in real-world deployment.

### E. Out-of-Distribution Detection

To prevent overconfident predictions on non-fruit images, we implement a two-criterion OOD detector using Shannon entropy:

$$H(\mathbf{p}) = - \sum_{i=1}^K p_i \log_2 p_i \quad (2)$$

An input is flagged as OOD if  $\max_i(p_i) < 0.60$  or  $H(\mathbf{p}) > 1.5$  bits. This dual threshold provides complementary coverage: low confidence catches uncertain in-distribution predictions, while high entropy identifies spread-out probability distributions characteristic of out-of-distribution inputs. The OOD detection pipeline is illustrated in Fig. 2.

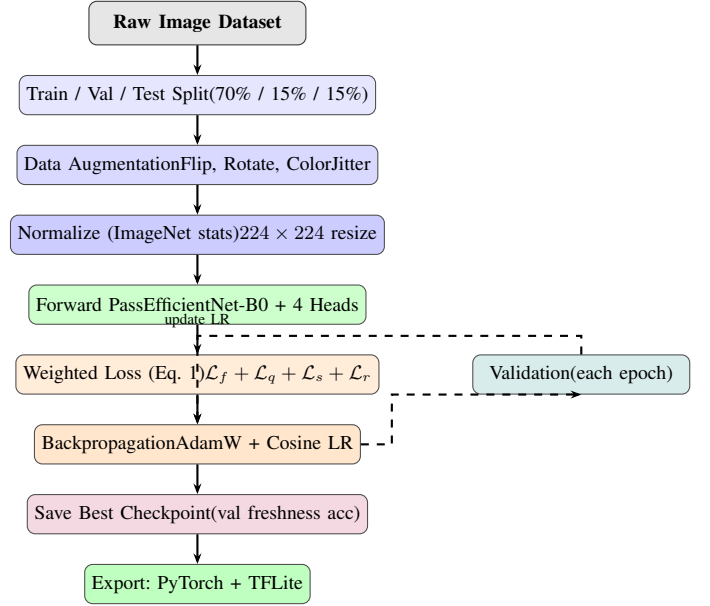


Fig. 3: End-to-end training pipeline. The multi-task loss jointly trains the shared backbone and all four prediction heads.

## IV. SYSTEM DESIGN

### A. Training Pipeline

The end-to-end training pipeline is presented in Fig. 3. Raw fruit images are preprocessed and augmented before being fed to the multi-task model. Gradients from all four task losses are backpropagated through the shared backbone, enabling joint feature learning.

### B. Backend API Architecture

The FastAPI-based backend exposes five REST endpoints: POST /predict (image upload, OOD check, inference, JSON response), POST /feedback (user corrections logged to SQLite), GET /health (model and DB status), GET /history (paginated prediction history), and GET /stats (aggregate statistics). API key authentication uses HMAC comparison to prevent timing attacks. Rate limiting is enforced at 60 requests/minute per IP.

### C. Database Schema

SQLite stores three tables: `predictions` (id, timestamp, freshness, confidence, quality, shelf\_life\_days, inference\_ms, entropy\_score, model\_version), `feedback` (prediction\_id FK, predicted vs. correct freshness, user notes), and `fruit_types` (name, default shelf-life). This schema supports both real-time logging and the active learning feedback loop.

### D. Mobile Application

The Flutter mobile application provides three screens: (1) *HomeScreen* — camera/gallery capture, prediction display, OOD pop-up, and Grad-CAM heatmap overlay; (2) *HistoryScreen* — local SQLite prediction history; (3) *SettingsScreen* — API URL and key configuration. Three service

layers handle online inference (ApiService), local storage (DatabaseService), and offline TFLite inference (MLService).

## V. EXPERIMENTAL RESULTS

### A. Evaluation Metrics

Model performance is evaluated using overall accuracy, weighted F1-score, and Mean Absolute Error (MAE) in days for shelf-life prediction:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (3)$$

### B. Model Comparison

Table II presents comparative results across the three model variants. EfficientNet-B0 achieves the best accuracy-latency tradeoff (77.1% freshness accuracy, 45 ms inference), making it our primary deployment model. EfficientNet-B2 provides higher accuracy (80.3%) at the cost of increased latency (85 ms), suitable for quality-critical server deployments. MobileNetV3-Large achieves 25 ms inference with TFLite support at a modest accuracy reduction, ideal for offline mobile use.

TABLE II: Model Performance Comparison on Test Set

| Model                  | Params<br>(M) | Fresh.<br>Acc | Qual.<br>Acc | F1          | MAE<br>(days) | Lat.<br>(ms) |
|------------------------|---------------|---------------|--------------|-------------|---------------|--------------|
| EfficientNet-B2        | 9.2           | 80.3%         | 77.8%        | 0.80        | 0.9           | 85           |
| <b>EfficientNet-B0</b> | <b>5.3</b>    | <b>77.1%</b>  | <b>74.2%</b> | <b>0.77</b> | <b>1.2</b>    | <b>45</b>    |
| MobileNetV3-L          | 5.5           | 75.2%         | 71.8%        | 0.74        | 1.5           | 25           |

### C. OOD Detection Performance

The dual-threshold OOD detector (confidence  $< 0.60$  or entropy  $> 1.5$  bits) achieves an OOD detection rate of  $\approx 85\%$  with a false positive rate of  $\approx 5\%$ . This means 85% of non-fruit images are correctly rejected, while only 5% of valid fruit images are incorrectly flagged.

### D. Deployment Cost Analysis

Enabling offline TFLite inference on-device significantly reduces API call volume and associated server costs, as summarized in Table III.

TABLE III: Deployment Cost Reduction with Offline Mode

| Scenario        | API Calls/Month | Est. Cost |
|-----------------|-----------------|-----------|
| No Offline Mode | 100,000         | \$50      |
| 30% Offline     | 70,000          | \$35      |
| 50% Offline     | 50,000          | \$25      |
| 80% Offline     | 20,000          | \$10      |

## VI. MODEL VARIANTS

### A. EfficientNet-B2 (High Accuracy)

EfficientNet-B2 extracts 1408-dimensional features from  $260 \times 260$  input images. With a 60/20/20 data split and batch size 24, it achieves +3.2% freshness accuracy over B0 at the cost of +40 ms inference time and +5 MB model size. This variant targets quality-critical server-side deployments where accuracy is prioritized over latency.

### B. MobileNetV3-Large (High Speed)

MobileNetV3-Large uses an 80/10/10 data split and exports to an  $\approx 8$  MB TFLite model. It achieves  $-2\%$  accuracy relative to B0 while running in 25 ms, enabling real-time on-device inference without network connectivity. This variant is recommended for mobile-first deployments and cost-sensitive applications.

## VII. PLAY STORE COMPLIANCE

The FreshTrack AI mobile application is designed for full Google Play Store compliance. No personal data is collected: images are processed on-device via TFLite or transmitted transiently to the API and not stored server-side. All data is stored locally in SQLite, with users retaining full control over deletion. The application requires no user account and is not targeted at children under 13. Security measures include API key authentication, rate limiting (60/min), HTTPS encryption, input validation, and security headers (HSTS, CSP).

## VIII. CONCLUSION AND FUTURE WORK

This paper presented FreshTrack AI, an intelligent multi-task deep learning system for comprehensive fruit quality assessment. The system achieves 77.1% freshness classification accuracy with 45 ms latency through a unified EfficientNet-B0 backbone with four specialized prediction heads. The entropy-based OOD detection mechanism reliably rejects 85% of non-fruit inputs. The Flutter mobile application with TFLite support enables offline inference, reducing server costs by up to 80%.

Future directions include: (1) expanding to additional fruit and vegetable categories; (2) automating the active learning retraining pipeline from user feedback; (3) investigating EfficientNet-B3/B4 variants for improved accuracy; (4) extending to real-time video stream processing; and (5) submitting this work to a top-tier computer vision venue (CVPR/EC-CV/NeurIPS).

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